python.assert#

https://pytorch.org/docs/stable/generated/exportdb/python.assert.html

Description:

Support Level: SUPPORTED Original source code:

Code Examples

```python
# mypy: allow-untyped-defs
import torch

class DynamicShapeAssert(torch.nn.Module):
    """
    A basic usage of python assertion.
    """

    def forward(self, x):
        # assertion with error message
        assert x.shape[0] > 2, f"{x.shape[0]} is greater than 2"
        # assertion without error message
        assert x.shape[0] > 1
        return x

example_args = (torch.randn(3, 2),)
tags = {"python.assert"}
model = DynamicShapeAssert()


torch.export.export(model, example_args)
```

```
ExportedProgram:
    class GraphModule(torch.nn.Module):
        def forward(self, x: "f32[3, 2]"):
            return (x,)

Graph signature:
    # inputs
    x: USER_INPUT

    # outputs
    x: USER_OUTPUT

Range constraints: {}
```

```
# mypy: allow-untyped-defs
import torch

class ListContains(torch.nn.Module):
    """
    List containment relation can be checked on a dynamic shape
or constants.
    """

    def forward(self, x):
        assert x.size(-1) in [6, 2]
        assert x.size(0) not in [4, 5, 6]
        assert "monkey" not in ["cow", "pig"]
        return x + x

example_args = (torch.randn(3, 2),)
tags = {"torch.dynamic-shape", "python.data-structure",
"python.assert"}
model = ListContains()


torch.export.export(model, example_args)
```

```
ExportedProgram:
    class GraphModule(torch.nn.Module):
        def forward(self, x: "f32[3, 2]"):
                 add: "f32[3, 2]" = torch.ops.aten.add.Tensor(x, x);  x = None
            return (add,)

Graph signature:
    # inputs
    x: USER_INPUT

    # outputs
    add: USER_OUTPUT

Range constraints: {}
```

Notes & Warnings

> **NOTE** Note Tags: python.assert Support Level: SUPPORTED

> **NOTE** Note Tags: torch.dynamic-shape, python.data-structure, python.assert Support Level: SUPPORTED

Detailed Documentation

dynamic_shape_assert#

Original source code:

list_contains#

Original source code:

Support Level: SUPPORTED

Original source code:

Tags: torch.dynamic-shape, python.data-structure, python.assert

Support Level: SUPPORTED

Original source code: